to calculate torque when the plane of the loop is parallel to the direction of magnetic field $\vec{B}$, i.e., $\hat{n} \perp \vec{B}$

$\hat{n} \perp \vec{B}$ **(unit normal vector is perpendicular to magnetic field)**

Since the current carrying wire experiences a force in a magnetic field, we shall tabulate the force experienced by each section of the loop and also the torque about an axis passing through the centre (see Table (A3.1))

$$\therefore \text{Net force } \vec{F}_{net} = \vec{F}_{PQ} + \vec{F}_{QR} + \vec{F}_{RS} + \vec{F}_{SP} = \vec{0}$$

And net torque

$$\vec{\tau}_{net} = \vec{\tau}_{PQ} + \vec{\tau}_{QR} + \vec{\tau}_{RS} + \vec{\tau}_{SP} = IabB\hat{j}$$

Thus, the net force on the rectangular loop is zero but net torque on the rectangular loop is not zero. Let A be the area of the rectangular loop (A = ab)

$$\text{Then } \vec{\tau}_{net} = ABI\hat{j}$$

If N be the number of turns of rectangular loop then

$$\vec{\tau}_{net} = NABI\hat{j}$$

Due to this torque, loop will start to rotate (here clockwise) and hence magnetic field $\vec{B}$ is no longer in the plane of the loop. Therefore, the above equation is a special case.

Note: When the plane of the loop is inclined to the direction of the magnetic field (i.e., $\hat{n} \not\perp \vec{B}$), then the torque is given by $\vec{\tau}_{net} = NABI\sin\theta\hat{j}$. In terms of magnetic dipole moment, $\tau_{net} = \vec{p}_m \times \vec{B}$.

Special Cases:

(i) $\theta = 90^\circ, \vec{\tau}_{net} = NABI\hat{j} = \max imum$

$\vec{p}_m and \vec{B}$ are perpendicular to each other

(ii) $\theta = 0^\circ, \vec{\tau}_{net} = \vec{0}$

$\vec{p}_m \; and \; \vec{B}$ are parallel

(iii) $\theta = 180^\circ, \vec{\tau}_{net} = \vec{0}$

$\vec{p}_m \; and \; \vec{B}$ are anti-parallel

EXAMPLE A3.1

Show the time period of oscillation when a bar magnet is kept in a uniform magnetic field is $T = 2\pi\sqrt{\frac{I}{p_m B}}$ in second, where I represents moment of inertia of the bar magnet, p_m is the magnetic moment and B is the magnetic field.

Solution

The magnitude of deflecting torque (the torque which makes the object rotate) acting on the bar magnet which will tend to align the bar magnet parallel to the direction of the uniform magnetic field $\vec{B}$ is

$$|\vec{\tau}| = p_m B\sin\theta$$

The magnitude of restoring torque acting on the bar magnet can be written as

$$|\vec{\tau}| = I\frac{d^2\theta}{dt^2}$$

Under equilibrium conditions, both magnitude of deflecting torque and restoring torque will be equal but act in the opposite directions, which means

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